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# Test-Time Refinement Moves Intelligibility Six Times More Than Identity in Masked-Diffusion TTS

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Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 A diffusion language model buys sequential computation twice: depth, paid once  
2 in parameters, and denoising steps, rented per utterance at serve time. We ask  
3 whether the rented kind helps every capability equally. We train an iso-parameter  
4 width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 20–125 M  
5 non-embedding parameters) and sweep the refinement budget  $T$  at inference only,  
6 giving a 75-point  $(w, d, T)$  surface per metric from 45 trained models. Fitting  
7  $\text{err} = E + Aw^{-\alpha} + Bd^{-\beta} + CT^{-\tau}$  pre-registered, test-time scaling is metric-  
8 selective:  $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$ , 95% run-level bootstrap CI [0.092, 0.131],  
9 excluding zero. The effect that carries practical weight is one of *magnitude*: over  
10  $T=1 \rightarrow 16$  refinement moves 0.952 absolute WER (a  $6.1\times$  reduction) against  
11 0.161 absolute speaker-similarity error ( $1.25\times$ ), a  $5.9\times$  asymmetry. It is not a  
12 difference in convergence *rate*: on each metric’s own normalised curve the fraction  
13 of total gain realised by  $T=8$  is 94.6% for WER and 93.7% for identity, identity  
14 marginally later. The selectivity persists across a 276 M four-budget replication,  
15  $\Delta\tau = +0.1095$ , CI [0.0951, 0.1241]. Depth and steps are *not* interchangeable:  
16 the substitution form  $B(dT^\kappa)^{-\beta}$  is worse than the separable form by  $\Delta\text{AICc} =$   
17  $+69.3$ .

## 18 1 Introduction

19 A diffusion language model can buy sequential computation twice. It *owns* depth:  $d$  transformer  
20 blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps:  
21  $T$  denoising passes over the same sequence, paid per utterance and adjustable at serve time. An  
22 autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the  
23 moment training ends. If rented serial compute substitutes for owned serial compute, a deployment  
24 can trade parameters for steps after the fact — but only for the capabilities that steps actually improve.

25 Zero-shot text-to-speech is an unusually clean probe of that question, because it separates two  
26 capabilities that are measured by standard, independent metrics. Intelligibility is compositional: the  
27 model must place the right phonetic content in the right order, and word error rate (WER) reads it  
28 out. Speaker identity is representational: the model must carry timbre from a short prompt into new  
29 content, and speaker similarity (SIM-o) reads that out. The two need not respond to the same knob.

30 We train one iso-parameter width–depth grid of masked-diffusion codec language models and then  
31 sweep  $T$  at inference only, which yields a full  $(w, d, T)$  response surface per metric at zero additional  
32 training cost. Two claims were pre-registered before any run. **H-D2**: refinement steps improve  
33 intelligibility more than identity, i.e. test-time scaling is metric-selective. **H-D3**: for intelligibility,  
34 steps and depth exchange at a measurable rate  $\kappa$  — how many step-doublings buy one depth-doubling.

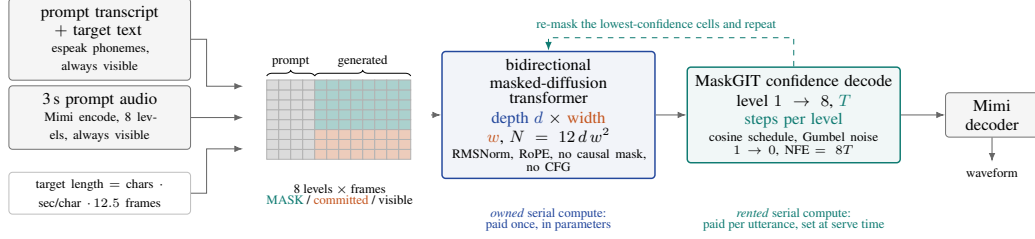


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes and a 3-second prompt are always visible; the generated region of the  $8 \times \text{frames}$  token grid starts fully masked. The backbone spends *depth* — *owned* serial compute, fixed at training time — and the sampler spends *T refinement steps per codebook level* — *rented* serial compute, chosen at serve time. One training grid over  $(w, d)$  therefore yields a whole  $(w, d, T)$  surface at inference. This paper asks whether the rented kind helps every capability equally.

$$\text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \quad \text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m (dT^{\kappa_m})^{-\beta_m} \quad (1)$$

Both claims were registered before the first run, together with the thresholds, the frozen sampler and the analysis plan. The answer to the first is yes and quantitative:  $\Delta\tau = 0.110$  with a run-level bootstrap CI of  $[0.092, 0.131]$ , so the step exponent is reliably larger for intelligibility than for identity. The answer to the second is a clean no: the substitution form is decisively *worse* than the separable one ( $\Delta\text{AICc} = +69.3$  for WER,  $+71.3$  for SIM-o) and  $\kappa$  runs to its bound, so no finite depth-step exchange rate describes this regime. The two results together say something sharper than either alone: extra inference compute is a targeted instrument, and it cannot be treated as a substitute for architecture.

- ① The first per-metric  $(w, d, T)$  scaling surface for speech generation: 75 points per metric from one training grid of 45 models, released with the harness.
- ②  $\Delta\tau = 0.110$ , CI  $[0.092, 0.131]$  — steps are metric-selective — and the *negative* result on depth-step substitution stated as an AICc comparison ( $+69.3$ ) rather than a narrative, with  $\kappa$  reported as unidentified.
- ③ A serving rule measured rather than assumed:  $c_{\text{layer}}$  is flat in *width* at batch 1 on the evaluation GPU, so serial latency is  $8Td_{\text{layer}}$  and the *step budget* is the only latency knob that trades against intelligibility.

## 2 Related work

Masked diffusion language models are now trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin et al., 2021], but their scaling is reported against a single likelihood-shaped objective, so a per-capability question cannot arise. In speech, the masked-generative recipes that made codec-token TTS practical — confidence-decoded parallel generation over residual codebook levels [Borsos et al., 2023, Chang et al., 2022] and its multi-stage successors [Wang et al., 2024] — fixed a step budget by engineering judgement and reported a single operating point; none maps error against steps as a law. Autoregressive codec TTS [Wang et al., 2023] has begun to report scaling behaviour, but in total parameters  $N$  only [Ye et al., 2025], which cannot separate width from depth and has no step axis at all; classical scaling work is likewise  $N$ -indexed [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2024], where the payoff is measured on one correctness metric; whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech. That is the gap this abstract fills: per-metric laws in  $(w, d, T)$  for a masked-diffusion TTS model.

## 3 Method

**Backbone and objective.** A bidirectional (non-causal) pre-LN transformer over Mimi [Défossez et al., 2024] codec tokens at 12.5 Hz with the first 8 residual codebooks (2048 entries each, plus

MASK and PAD). Width  $w$ , depth  $d$ , head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE, no weight tying, so the non-embedding parameter count is  $N = 12dw^2$ . One embedding table per codebook is summed at the input and one linear head per codebook reads out. Training is SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example a level  $l \sim \mathcal{U}\{1..8\}$  and a ratio  $t \sim \mathcal{U}(0, 1]$ ; levels below  $l$  stay visible, level  $l$  is masked i.i.d. with probability  $t$ , levels above  $l$  are entirely MASK; the loss is cross-entropy on the masked cells of level  $l$  only. A phoneme prefix and a 3-second prompt of codec tokens are always visible; there is no speaker encoder, so the backbone itself is the only identity pathway, and no classifier-free guidance, which would add a second inference-time axis and confound the step axis. The target length is fixed before generation from the text length and a corpus character rate (0.0602 s/char).

**Frozen sampler.** MaskGIT-style confidence decoding [Chang et al., 2022], level by level, with exactly  $T$  refinement steps per level, so NFE =  $8T$ : a cosine schedule sets how many cells remain masked after each step, tokens are drawn categorically at temperature 1.0, and confidences are log-probabilities plus Gumbel noise annealed  $1 \rightarrow 0$  within a level; committed cells stay committed. The sampler is identical for every config, seed and  $T$  — nothing is tuned per shape or per step budget. Per-item generation noise is keyed so that each item sees the same noise at the first step of every level for every  $T$ , making  $T$  the only difference between surface points of one run. Before any metric is computed we verify that  $T$  reaches the sampler: outputs at  $T=1$  and  $T=16$  must differ on more than 20% of generated token cells.

**Design and statistics.** 15 configs span three iso- $N$  budgets  $\times$  five aspect ratios, each trained at 3 seeds for 30,000 steps at effective batch 256 with  $\mu P$  [Yang et al., 2021] (base width 256, base LR 0.004); width transfer and depth transfer were verified by five-point LR sweeps before the grid.  $T \in \{1, 2, 4, 8, 16\}$  is swept at evaluation only, giving 75 surface points per metric from 45 trained models. Metrics: WER (Whisper-large-v3 [Radford et al., 2023], greedy, the Whisper English normaliser), SIM-o (WavLM-large [Chen et al., 2022] speaker-verification embedding against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-output rule, and UTMOS [Saeki et al., 2022] as a tertiary. Error transforms are  $\text{err}_{\text{WER}} = \text{WER}$  and  $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$ , computed over all items including degenerates. Both forms of Eq. (1) are fitted by weighted non-linear least squares with 32 multi-starts;  $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$ ,  $\kappa$ , and  $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$  carry percentile confidence intervals from a bootstrap that resamples at the *run* level — the five  $T$ -points of one run share a trained model, so resampling surface points would understate their correlation — with 2000 replicates recomputing every  $T$  point.

## 4 Results

**Steps are metric-selective (H-D2, supported).** Fitted on all 75 surface points, the step exponent is  $\tau_{\text{WER}} = 0.836$  (CI [0.825, 0.851]) against  $\tau_{\text{SIM}} = 0.726$  (CI [0.709, 0.744]), so  $\Delta\tau = 0.110$  with CI [0.092, 0.131], excluding zero. The raw surface shows it without any model: over the grid,  $T=1 \rightarrow T=16$  takes WER from 113.8% to 18.7% (a factor of  $6.1\times$  on the error scale) while SIM-o moves only  $0.209 \rightarrow 0.370$  ( $1.25\times$  on  $1 - \text{SIM-o}$ ), an absolute-error asymmetry of  $5.9\times$ . The selectivity is not an artefact of scale, training compute, or the metric stack: extending the grid to a fourth budget (276 M non-embedding parameters, ten further runs) gives  $\Delta\tau = +0.1095$ , CI [0.0951, 0.1241]; retraining three configurations for  $3\times$  longer gives  $\Delta\tau = +0.1767$ , CI [0.0789, 0.2854]; and re-scoring every synthesis under a different ASR or refitting with unbounded log-amplitudes moves it to  $+0.1283$  and  $+0.1260$  respectively. Extending  $T$  to 64 on a 21-run subset shows WER still falling past the grid edge ( $0.1516 \rightarrow 0.1294$ ), so  $T_{\text{WER}}^*$  on WER’s own scale is 32, confirming the  $T=16$  value was a lower bound. The power gate passed in all three budgets before these tests were run.

**Depth and steps do not substitute (H-D3, not supported).** The pre-registered substitution form collapses:  $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$  for WER and  $+71.3$  for SIM-o, where the hypothesis required  $\leq +4$ , and the fitted  $\kappa$  sits on its bound for both metrics, i.e. the exchange rate is not identified. We therefore quote no  $\kappa$ . The mechanism is visible in the fits:  $T$  enters with a large additive term that decays as  $T^{-\tau}$  towards a floor, whereas depth changes the floor itself; a model that forces the two into one factor  $dT^\kappa$  cannot fit both the strong depth ordering at fixed  $T$  and the strong  $T$  ordering at fixed depth.

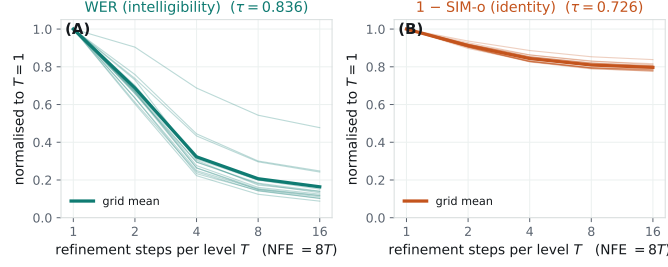


Figure 2: *Test-time scaling is metric-selective.* Error versus refinement steps for all 15 configs, each normalised to its own  $T=1$  value; thick lines are grid means. (A) WER error falls 0.9518 in absolute terms over  $T=1 \rightarrow 16$  and is still falling at  $T=16$ . (B) SIM-o error falls 0.1606 over the same range,  $5.9\times$  less. Same models, same frozen sampler, same items — only  $T$  differs.

Table 1: *Fitted exponents* on the 75-point surface ( $M_{\text{sep}}$  for  $\alpha, \beta, \tau$ ;  $M_{\text{sub}}$  for  $\kappa$ ) with 95% run-level bootstrap CIs for the step exponents.  $\Delta\tau = 0.110$ , CI [0.092, 0.131], excludes zero: the pre-registered primary result.  $\kappa$  is at its bound in both rows and  $\Delta\text{AIC}_{\text{csub-sep}} \gg +4$ , so no exchange rate is quotable.  $\Delta\text{AIC}_{\text{cfull-N}} \leq -4$  means iso- $N$  shape matters.

| Metric | $\alpha$ (width) | $\beta$ (depth) | $\tau$ (steps)              | $\kappa$           | $\Delta\text{AIC}_{\text{csub-sep}}$ | $\Delta\text{AIC}_{\text{cfull-N}}$ |
|--------|------------------|-----------------|-----------------------------|--------------------|--------------------------------------|-------------------------------------|
| WER    | 1.574            | 2.040           | <b>0.836</b> [0.825, 0.851] | 3.000 <sup>†</sup> | +69.3                                | −15.5                               |
| SIM-o  | 0.919            | 0.553           | <b>0.726</b> [0.709, 0.744] | 3.000 <sup>†</sup> | +71.3                                | −5.5                                |

<sup>†</sup>at the fitted bound, i.e. not identified;  $\Delta\tau = 0.110$ , CI [0.092, 0.131].

**Shape matters, but the anisotropy ratio is not identified (H-D1, not supported).** At  $T=16$  the two-axis model beats the  $N$ -only model for both metrics ( $\Delta\text{AICc} = -15.5$  for WER,  $-5.5$  for SIM-o, against a  $-4$  threshold), so error is not a function of parameter count alone — iso- $N$  shape choice matters. The direction of the anisotropy, however, we cannot claim:  $\Delta\rho = 0.174$  with CI  $[-0.711, 0.388]$ , spanning zero, and the width amplitude  $A$  saturates the pre-registered bound  $A \leq 10$  in every fit, so only  $A w^{-\alpha}$  is identified and  $\alpha$  — hence  $\rho = \alpha/\beta$  — is not. The **depth** exponent, by contrast, *is* identified ( $B$  interior in both fits) and separates the capabilities sharply:  $\beta_{\text{WER}} = 2.040$  against  $\beta_{\text{SIM}} = 0.553$ . What the data do show unambiguously is a strong **depth** ordering at fixed parameter count: moving from the shallowest shape to the best-performing depth is worth 40.2, 16.3 and 7.3 WER points in the 20, 50 and 125 M budgets, while SIM-o varies by under 0.065 across all 15 shapes. The depth optimum is *interior* only in the smallest budget ( $d=18$  of 4–24); in the two larger budgets the argmin sits at the deepest shape tested ( $d=30$  of 6–30 and  $d=36$  of 8–36), so those are censored lower bounds and we claim no  $d^*(N)$  trend.

**Extrapolation and reliability.** Fitting the two smaller budgets and predicting the largest gives MAPE 2.4% for SIM-o (versus 2.2% for the  $N$ -only model), inside the 15% the hypothesis asked for; WER does not extrapolate (20.2% versus 20.4%). Degenerate outputs — empty, truncated or looping transcripts under the frozen rule — fall monotonically with the step budget, from 40.4% at  $T=1$  to 0.1% at  $T=16$ : most of what  $T=1$  costs is not subtle quality but outright failures.

**Serving corollary.** Measured on the evaluation GPU at batch 1,  $c_{\text{layer}}$  is flat in width across the whole grid (0.493–0.514 ms per layer for  $w \in [256, 1152]$ ): at these sizes the device is launch-bound, so serial latency is  $8Td c_{\text{layer}}$  and width is free, so a latency budget constrains only the product  $Td$ . These data do not resolve how to split it: the pre-registered rule that depth should be saturated before steps are raised failed (satisfied at 5.9% of 40 log-spaced budgets by the measured method and 8.8% by the fitted one, against an 80% bar), and  $T$  is censored at 16 by the tested grid, so no directional allocation rule is claimed in either direction.

**Where the steps are spent matters as much as how many (exploratory).** At matched total NFE = 32 on the 125 M budget, spending every refinement pass on the coarsest codebook level rather than the finest changes WER from 1.1480 to 0.2191 (paired item-level bootstrap difference  $-0.9289$ , CI  $[-0.9595, -0.8999]$ ) and the degenerate rate from 39.1% to 0.1%; the coarse schedule is

not free, costing 0.3057 against 0.3477 SIM-o and 2.17 against 2.51 UTMOS relative to a uniform split (Appendix C).

## 5 Limitations and outlook

The ceiling is 133 M non-embedding parameters and 30,000 steps; every model is under-trained by frontier standards, and the  $T$  curves may flatten differently at scale —  $T_{\text{WER}}^* = 16$  on WER’s own scale is a lower bound, since WER had not saturated at the largest budget we measured. One language, one codec, one corpus, and no classifier-free guidance — guidance is a second inference-time axis that would confound the step axis, so excluding it is deliberate, and it is the obvious follow-up. The  $\Delta\tau$  interval resamples 3 runs per config, so it captures seed-to-seed variation only, not the uncertainty of the single shared harness and corpus. Target length is set from a corpus-median seconds-per-character rather than per-item speaking rate, so an utterance whose speaker is unusually fast or slow is synthesised at the wrong duration; this penalises both metrics and is not corrected for.  $\Delta\rho$  and  $\kappa$  are reported as unidentified rather than estimated. Cuts taken: none. This is pre-registered outcome class S1, and selectivity is the only positive claim the data license.

**Reproducibility.** Protocol, hypotheses, thresholds, frozen sampler and analysis plan were registered before the first run; every gate is logged with its measured values. The released artefact holds the harness, all 45 run records, the 75-point surface, and the fitting and bootstrap code. Total cost 143 GPU-hours.

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## 222 A The shape grid, as run

223 Every config in the active grid with its measured  $T=16$  means (seeds 0, 1, 2, 400 eval\_zs items each).  
224 Iso- $N$  within a budget: the three budgets sit at 20, 50 and 125 M non-embedding parameters and the  
225 five shapes per budget span the aspect-ratio range at fixed  $N = 12 d w^2$ .

| config | budget | $w$  | $d$ | heads | $N$ (M) | val loss | WER (%) | SIM-o | Degen (%) |
|--------|--------|------|-----|-------|---------|----------|---------|-------|-----------|
| A1     | A      | 640  | 4   | 10    | 19.7    | 4.638    | 53.9    | 0.327 | 0.67      |
| A2     | A      | 448  | 8   | 7     | 19.3    | 4.582    | 28.4    | 0.348 | 0.08      |
| A3     | A      | 384  | 12  | 6     | 21.2    | 4.566    | 15.9    | 0.360 | 0.17      |
| A4     | A      | 320  | 18  | 5     | 22.1    | 4.574    | 13.6    | 0.359 | 0.00      |
| A5     | A      | 256  | 24  | 4     | 18.9    | 4.611    | 15.3    | 0.351 | 0.17      |
| B1     | B      | 832  | 6   | 13    | 49.8    | 4.551    | 27.8    | 0.361 | 0.08      |
| B2     | B      | 640  | 10  | 10    | 49.2    | 4.505    | 16.0    | 0.381 | 0.00      |
| B3     | B      | 512  | 16  | 8     | 50.3    | 4.497    | 11.9    | 0.388 | 0.08      |
| B4     | B      | 448  | 22  | 7     | 53.0    | 4.511    | 12.0    | 0.382 | 0.08      |
| B5     | B      | 384  | 30  | 6     | 53.1    | 4.520    | 11.5    | 0.375 | 0.08      |
| C1     | C      | 1152 | 8   | 18    | 127.4   | 4.504    | 17.1    | 0.373 | 0.17      |
| C2     | C      | 960  | 12  | 15    | 132.7   | 4.504    | 18.9    | 0.373 | 0.17      |
| C3     | C      | 768  | 18  | 12    | 127.4   | 4.488    | 14.3    | 0.392 | 0.00      |
| C4     | C      | 640  | 26  | 10    | 127.8   | 4.489    | 13.4    | 0.390 | 0.08      |
| C5     | C      | 512  | 36  | 8     | 113.2   | 4.487    | 9.9     | 0.388 | 0.00      |

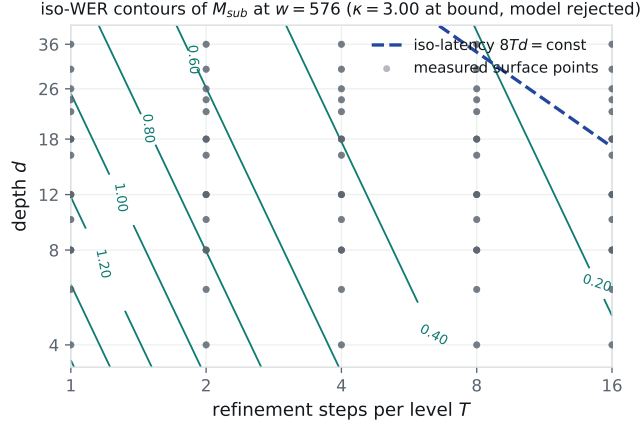


Figure 3: *Why the substitution form is rejected.* Iso-WER contours of  $M_{\text{sub}}$  in  $(\log T, \log d)$  at the median width, with the iso-latency line  $8Td = \text{const}$ . The fitted  $\kappa$  sits on its bound and  $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ , so these contours are shown as a rejected model, not as a design rule.

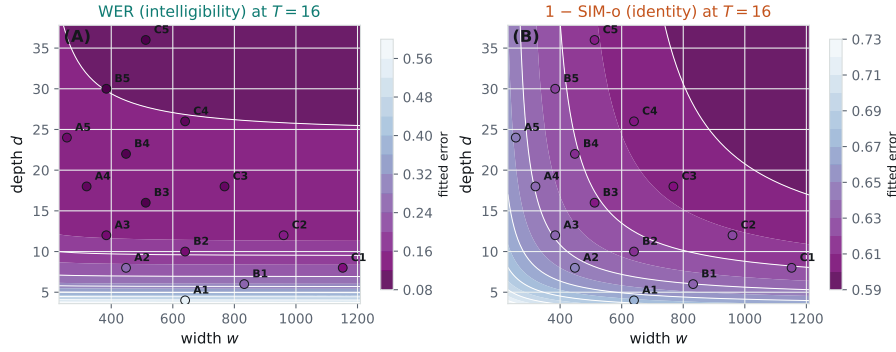


Figure 4: *Fitted anisotropy surface at  $T=16$  for both metrics, with the measured config means overlaid.* The width amplitude  $A$  saturates its pre-registered upper bound for both metrics, which is why  $\rho = \alpha/\beta$  is reported as unidentified in §4 even though the two-axis model beats the  $N$ -only model ( $\Delta\text{AICc} = -15.5$  and  $-5.5$ ).

## 227 B Additional figures

## 228 C Refinement-budget allocation across codebook levels

229 Three frozen per-level step schedules at matched total NFE = 32 on the 125 M budget (15 runs, 400  
 230 items). Paired item-level bootstrap, 2000 replicates, pairing on the eval item so item difficulty cancels.  
 231 Coarse spends all passes on codebook level 0, fine on level 7, uniform splits evenly.

| schedule               | WER    | SIM-o  | UTMOS | degenerate |
|------------------------|--------|--------|-------|------------|
| coarse [25, 1, ..., 1] | 0.2191 | 0.3057 | 2.17  | 0.1%       |
| uniform [4, ..., 4]    | 0.3144 | 0.3477 | 2.51  | 0.5%       |
| fine [1, ..., 1, 25]   | 1.1480 | 0.2246 | 1.56  | 39.1%      |

233 Allocation is metric-selective in the same way the step *count* is: coarse wins WER but loses identity  
 234 and naturalness against uniform, both with intervals excluding zero. It is a trade, not a free win.

235 **A significance-based saturation contrast.** Defining saturation as the smallest  $T$  whose paired-  
 236 bootstrap interval for  $\text{err}(T) - \text{err}(64)$  covers zero gives  $T_{\text{WER}}^* = 32$  against  $T_{\text{SIM}}^* = 16$  on the  
 237 extended grid. We report this only here and do not build on it: roughly half the contrast is a precision  
 238 artefact, since identity’s detection floor is  $\approx 1.8\times$  looser relative to its own dynamic range than

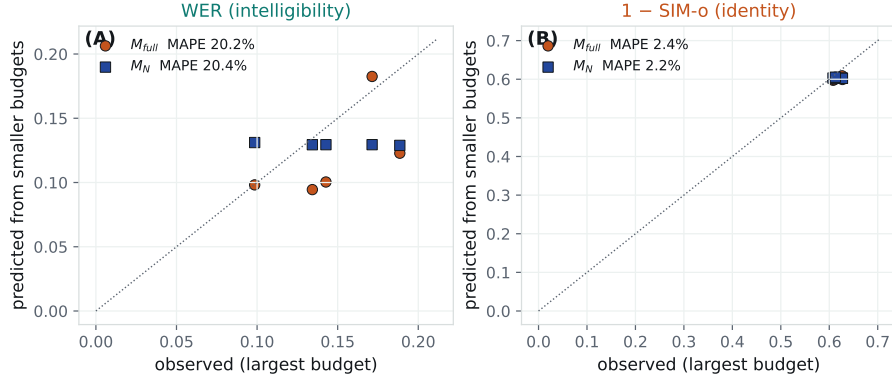


Figure 5: *H-D4 extrapolation*. The two smaller budgets are fitted and the largest budget predicted, under  $M_{full}$  and the  $N$ -only model. SIM-o extrapolates (MAPE 2.4 % vs 2.2 %); WER does not (20.2 % vs 20.4 %).

intelligibility's. The raw-scale  $T^*$  — 95% of the  $T=16$  value — is not affine-invariant and must never be compared across metrics: the same 5% band is 0.67% of WER's range but 11.96% of SIM-o's. On the invariant fraction-of-total-gain definition,  $T^* = 16$  for both metrics.

## D Fit diagnostics and exploratory analyses

**Parameters at bounds.** The width amplitude  $A$  saturates the pre-registered *upper* bound  $A \leq 10$  in every fit (Part A: 10.000 for both metrics; Part B: 9.999 for WER, 10.000 for SIM-o), so only the product  $A w^{-\alpha}$  is identified and  $\alpha$  is not; the bound, not the data, is binding, which is a limitation of the pre-registered parameterisation worth fixing in any follow-up.  $\kappa$  reaches its upper bound in every one of the 2000 bootstrap replicates of  $M_{sub}$ . The irreducible term  $E$  reaches 0 for WER under  $M_{sep}$ . The depth amplitude  $B$  is interior throughout, so  $\beta$  is identified.

**Exploratory: the depth exponent separates the capabilities.**  $\beta_{WER} = 2.040$  against  $\beta_{SIM} = 0.553$ , a factor of 3.7 — the asymmetry H-D1 sought, in the parameter the data pin down. Part-B  $\rho$  values point the pre-registered way ( $\Delta\rho = +0.890$ ) but inherit the  $A$ -cap problem; the pre-registered test is Part A, so this is exploratory only.

**Weighting.** Points are weighted  $1/SE^2$  with SE taken over seeds, floored at a quarter of the pooled seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored points is recorded in the released `fits.json`.

**Exploratory (not part of any declared claim).** An unweighted refit of both parts is released alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed. Both reproduce the sign and significance of  $\Delta\tau$ .

## Use of Generative AI Disclosure

Generative AI was used as a coding assistant for the experimental harness and for language refinement of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and take full responsibility for this paper.